

DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE

Bachelor of Technology (Artificial Intelligence and Data Science) SEMESTER - 6 Summer 2025 (Regular)

Course :Bachelor of Technology (Artificial Intelligence and Data Science) Branch : Engineering and Technology

Semester : SEMESTER - 6

Subject Code & Name: BTAIPE603D - WEB DEVELOPMENT

Time : 3 Hours]

[Total Marks : 60

Instructions to the Students:

1. Each question carries 12 marks.
2. Question No. 1 will be compulsory and include objective-type questions.
3. Candidates are required to attempt any four questions from Question No. 2 to Question No.6
4. Use of non-programmable scientific calculators is allowed.
5. Assume suitable data wherever necessary and mention it clearly.

Q1. Objective type questions. (Compulsory Question)

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- 1 What is Node.js primarily used for?
 - a) Styling web pages
 - b) Running JavaScript on the server
 - c) Designing databases
 - d) Writing HTML code
- 2 Which of the following is a component of the Struts framework?
 - a) ServletContainer
 - b) Action class
 - c) JSPBean
 - d) SpringBoot
- 3 Which of the following is used to validate the structure of an XML document?
 - a) HTML
 - b) JavaScript
 - c) DTD
 - d) JSON
- 4 Which DOM method is used to get an element by its ID?
 - a) getElementByClass()
 - b) getElementById()
 - c) querySelectorAll()
 - d) findById()
- 5 Which tag is used to create a hyperlink in HTML?
 - a) <a>
 - b) <link>
 - c) <href>
 - d) <url>
- 6 Which lifecycle method is called when a servlet is being removed from memory?
 - a) init()
 - b) doGet()
 - c) service()
 - d) destroy()
- 7 Which component in Struts is responsible for handling the client request?
 - a) View
 - b) Action class
 - c) Interceptor
 - d) Controller Servlet
- 8 In XML, a declaration is used at the beginning of every well-formed document.
 - a) `<?xml version="1.0" ?>`
 - b) `<!DOCTYPE>`
 - c) `<!-- comment -->`
 - d) `<xml>`
- 9 Which of the following is a valid way to define a function in JavaScript?
 - a) `function = myFunc()`
 - b) `function myFunc() { }`
 - c) `function:myFunc()`
 - d) `def myFunc()`
- 10 Which method is called only once during the lifecycle of a servlet?
 - a) doGet()
 - b) doPost()
 - c) init()
 - d) service()
- 11 Which of the following is used to describe a web service?
 - a) UDDI
 - b) WSDL
 - c) XML
 - d) SOAPUI
- 12 PHP global variable retrieves data sent with a form using POST method
 - a) `$_FORM`
 - b) `$_POST`
 - c) `$_GET`
 - d) `$_REQUEST`

Q2. Solve the following.

- A) Describe the structure of an HTTP request message. Create a simple HTML page with a heading, a paragraph, and an image. 6
- B) Design and implement a complete HTML form for user registration that includes the appropriate fields: Full Name, Email Address, Password, Gender, Date of Birth, Phone Number, Address, City, Interests, and Submit and reset Buttons. 6

Q3. Solve the following.

- A) Distinguish between primitive and reference data types in JavaScript. Explain the difference between var, let, and const in JavaScript with a suitable example. 6
- B) Write a JavaScript function to find the largest number among three numbers entered by the user and return the result. 6

Q4. Solve Any Two of the following.

- A) Describe the servlet architecture and how it processes client requests. Discuss how parameter data is received and processed in servlets with example. 6
- B) Describe the purpose of XML namespaces with a suitable example. Define DTD, discuss the type of DTD, give an example of any one of its types, and explain the role of DTD in XML validation. 6
- C) What is the purpose of *web.xml* in a servlet project? Name two XML schema definition types. Write an XML document to represent a student record that includes: student ID (as an attribute), name, age, gender, course, year of enrollment, email, and address (with nested elements for street, city, and postal code). Ensure the XML is well-formed and structured. 6

Q5. Solve Any Two of the following.

- A) Explain the relationship between JSP and Servlets. Describe the process of running a JSP application. 6
- B) Write a client program to consume a Java web service using HTTP to call the following webpage: "http://localhost:8090/MyWebService/ApplicationForm" and return plain text or JSON data. 6
- C) Implement form validation in a Struts web application using the built-in Struts Validation Framework. Create a user registration form with fields: username, password, email, and age. Apply the validation rules and ensure meaningful error messages are displayed when validation fails. Use *validation.xml* or annotations for defining validation rules. 6

Q6. Solve Any Two of the following.

- A) Explain the basic syntax and structure of a PHP script. Describe how PHP handles form data using *\$_POST* and *\$_GET*. 6
- B) Explain the steps in establishing a connection to a MySQL database using PHP, and illustrate with appropriate syntax. 6
- C) What is the *.NET* framework? Define a web service in *ASP.NET*. Write a simple C# program to demonstrate control flow using *if-else* and loops. 6

*** End ***